Personal Reading Report: SlowFast Networks for Video Recognition

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Role: 3D Spatio-Temporal Network Implementation
Paper Assessed: *SlowFast Networks for Video Recognition* (Feichtenhofer et al., ICCV 2019)

1. Overview & Problem Addressed

In video modeling, there is a fundamental asymmetry between space and time: spatial semantics (colors, textures, objects) evolve slowly, while temporal dynamics (motion, actions) can change extremely rapidly. Traditional 3D CNNs treat spatial and temporal dimensions symmetrically, which is computationally expensive and physically unintuitive.

The SlowFast paper addresses this by proposing a dual-pathway architecture inspired by the biological visual system of primates (P-cells and M-cells), completely avoiding the need for computationally heavy Optical Flow pre-processing.

2. Core Architectural Concepts

Through my literature review, I have identified three critical components of the SlowFast architecture that we need to consider for our implementation:

- **The Slow Pathway:**
  - Operates at a low frame rate (large temporal stride $\tau = 16$).
  - Maintains a large channel capacity (C) to capture detailed spatial semantics.
  - Consumes ~80% of the total computational FLOPs.
- **The Fast Pathway:**
  - Operates at a high temporal resolution (refresh rate is $\alpha = 8$ times higher than the Slow pathway).
  - Crucially, it is lightweight. The channel capacity is reduced to a fraction ($\beta = 1/8$), which makes it highly efficient. Because 3D convolution cost scales quadratically with channels O(C^2)), this pathway only consumes ~20% of total FLOPs despite processing 8x more frames.
- **Lateral Connections (Feature Fusion):**
  - Information flows unidirectionally from Fast to Slow.
  - To match the mismatched temporal dimensions between the two pathways, the authors use a **Time-to-Channel (T2C)** reshape technique. By folding the temporal dimension ($\alpha = 8$) into the channel dimension ($\beta = 1/8$), the product is $\alpha\beta = 1$, perfectly matching the channel dimension C of the Slow pathway.

3. Relevance to CS-412 Project (2D vs 3D Benchmark)

For our benchmark project (*Real-Time Driver Distraction Detection: A 2D vs. 3D Vision Benchmark*), SlowFast will serve as the **3D Spatio-Temporal Baseline**.

- **Strengths for DMS (Driver Monitoring System):** It is expected to excel at recognizing continuous actions (e.g., reaching behind, answering a phone) where temporal context is crucial, potentially outperforming the 2D CNN/ViT-DD model (assigned to Ky An).
- **Challenges for DMS:** 3D CNNs traditionally suffer from higher inference latency. Our evaluation will focus on whether the lightweight Fast pathway allows SlowFast to maintain an acceptable FPS for real-time edge deployment compared to Frame-level approaches.